Authority Aware and Evidence Verified RAG for Reliable Legal Research over Indian Judicial Judgements

Tharuneesh GK
M.Tech Data Science
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India
Tharuneeshgk24@gmail.com

Mrs. Leena Jenifer
Department of IT
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India
Leenajenifer.l@rajalakshmi.edu.in

Dr. Ganesa Ramachandran Arunachalam
L&T EduTech, Chennai
grcchem@gmail.com

Abstract—Finding a relevant judgment does not guarantee that a legal retrieval system will read the passage needed to check a claim. This paper studies the issue in LEXRAG by comparing five evidence selection policies. The experiment includes an opening 1000 character prefix, the current window policy, and three selectors limited to 380 evidence tokens. Tests cover 54 synthetic examples and 30 constructed claims from six unmodified Indian judgment texts. In the real text pilot, agreement with frozen AI assisted draft labels rises from 40.0% with the character prefix to 53.3% with current windows. Both semantic and uniform selection reach 50.0% at the same token budget, yet no method identifies any of the 12 draft supported claims, even when the annotated evidence is recovered. The results demonstrate the need to separately assess passage coverage and quality of inference. This is a reproducible diagnostic procedure which requires independent annotation and new test cases to demonstrate legal accuracy.

Index Terms— legal information retrieval; evidence selection; natural language inference; claim verification; Indian judgments; evaluation

I. INTRODUCTION

A legal research tool must do more than return a judgment on a similar topic. It must also locate the passage that answers the user's question and identify whose statement that passage records. An allegation, a witness account, and a court finding may describe the same event without carrying the same meaning. This distinction motivates separate measures for retrieval and verification [1], [2]. Work on Indian legal question answering also shows the need to combine automated evaluation with domain informed human assessment [3].

Long judgments create a practical constraint: the verifier can inspect a limited amount of text at a time. Taking the opening passage is simple but may omit later findings. Reading multiple windows increases the chance of seeing that finding but also introduces more text and potentially conflicting statements. A related prefix based NLI design is discussed in Section II. Here we examine the selection step in LEXRAG, asking whether any improvement remains when budgets are matched and whether the model interprets the passages it receives correctly.

We compare five selectors using frozen sources and report passage recovery alongside classification outcomes. A synthetic position test isolates omissions due to prefix selection. A small pilot on real judgments reveals how the verifier behaves on less controlled text. The pilot annotations were drafted by an assistant and have not received independent legal review, so its scores represent draft label agreement. The contribution is a reproducible investigation of these implementation issues, including a failure to recognize supported claims. It is not a claim that window selection is new.

II. LITERATURE REVIEW

The evaluation literature provides two useful starting points. LegalBench RAG focuses on precise evidence spans annotated by experts [1], while LRAGE separates components of a legal RAG pipeline to examine individually [2]. AILQA adds Indian legal context and expert assessment alongside automatic measures [3]. These motivate our decision to measure evidence recovery separately from the verifier's final label.

Other work examines what makes a judgment useful as a retrieved case. CBR RAG explores case representations and different forms of similarity [4]. Precedent Aware Multi Agent RAG proposes considering jurisdiction, authority, temporal validity, and negative treatment [5]. LegalGraphRAG uses a hierarchical legal graph and separate agents to find and assess evidence [6]. These extend beyond textual entailment: a case can resemble the current dispute without establishing facts or providing binding authority.

Falkor IRAC links legal reasoning to supporting graph paths and identifies doctrinal conflicts in an Indian judicial proof of concept [7]. Kumar and Dhir combine evidence grounding with evaluation and template based repair [8]. Both provide alternatives for checking or revising legal outputs. Our experiment addresses an earlier and narrower question: how choice of passages affects inference given a fixed model.

Document structure offers another avenue to improve retrieval. Rhetorical Role Aware RAG uses legal roles along with lexical retrieval, dense retrieval, fusion and reranking [9]. AQgR retrieves Indian cases through generated questions and structured summaries [10]. Domain Partitioned Hybrid RAG routes queries across different legal sources and includes graph information [11]. These suggest useful future selectors, particularly for separating facts and judicial reasoning, but each requires evidence level evaluation.

Wicaksono et al. compare retrieval augmented legal question answering methods on Indonesian human trafficking decisions [12]. Their study supports retaining a lexical baseline, although their findings cannot simply be transferred to a different jurisdiction and task. NyayaRAG uses statutes and precedents for Indian judgment prediction [13], and NyayaMind develops structured reasoning for that setting [14]. Our task differs: checking a proposition against a supplied source rather than predicting a decision in a new case.

LegalMind is the closest motivation for the prefix comparison because its architecture includes NLI verification alongside retrieval, a tuned language model, and a Judge agent [15]. We evaluate a MiniLM verifier in LEXRAG, not a reproduction of that complete system. Naveen et al. explore legal argumentation through a neuro symbolic multi agent design [16], and NyayaAI coordinates several legal assistance tasks through specialized agents [17]. These broader workflows are distinct from the passage selection problem tested here.

The remaining application studies illuminate different deployment priorities. Legal Gennie emphasizes multilingual interaction and document aware legal assistance [18]. The transformer retrieval framework of Karthik Surya et al. provides a lightweight legal RAG design [19]. Vakil combines domain adaptation with retrieval and graph related components [20]. These may improve access or system capability, but their benefits should be tested separately from claim verification accuracy.

For LEXRAG, the unresolved question is whether windows help because they select better passages or are simply exposing the model to more text. A second question is whether the verifier can use the passage correctly once it is visible. We address both by using equal evidence budget comparisons, span recovery, and error analysis, and the production window policy. This is a specific gap in the evaluated implementation. The supplied literature motivates the experiment but is not an exhaustive basis for claiming prior research has overlooked the problem.

III. PROPOSED ARCHITECTURE AND METHODOLOGY

A. System Architecture and Experimental Boundary

The deployed LEXRAG retrieval path combines BM25 and all MiniLM L6 v2 embeddings through reciprocal rank fusion, followed by an MS MARCO MiniLM cross encoder reranker. A separate claim checker uses cross encoder nli MiniLM2 L6 H768. The inspected CSV contains 7030 rows with Text and Summary fields; the active index contains 6000 chunks from 286 judgments. The CSV lacks court and date metadata, so this configuration cannot substantiate authority aware ranking. The local system retrieves and checks claims; it is not a generative question answering model.

To isolate selection, the experiment supplies the same complete frozen judgment to every selector for a claim. It bypasses retrieval, the ingestion chunk cap, and the optional claim focused retry. Consequently, results describe verification conditional on the supplied source. They do not measure retrieval recall, generated answer quality, or an end to end legal research workflow

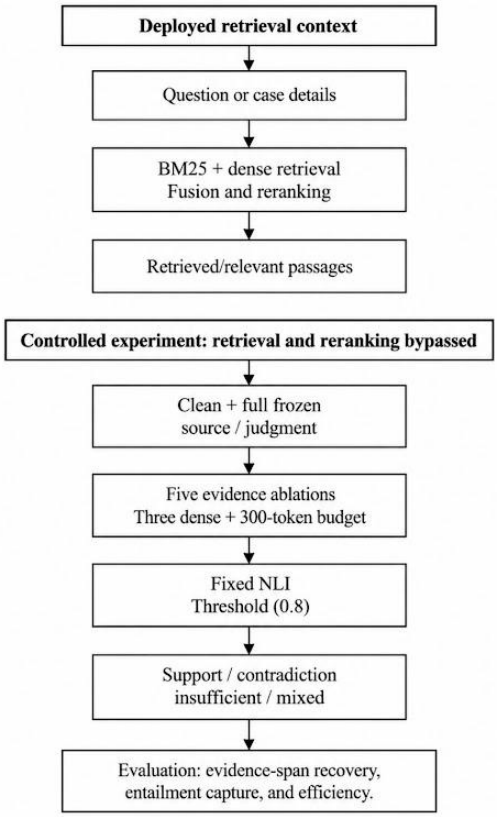


Fig. 1. Retrieval context and the isolated evidence selection experiment. The lower path is the evaluated system.

B. Label Meaning and Claim Construction

Supported means that the supplied evidence establishes the exact claim. Contradicted means that it establishes an incompatible statement about the same relevant people, event, and time. Insufficient evidence means that the supplied material establishes neither. These labels concern the source record, not truth outside it. Mixed is a system diagnostic for simultaneous strong support and contradiction; it is not a fourth reference class.

For example, if a judgment states that Revai repaid Meena, the claim that this same debt remained unpaid is contradicted, provided the event and time match. If an unrelated earlier case states that another borrower repaid a debt, that judgment may be relevant precedent but does not establish whether Revai paid Meena. The current factual claim is then insufficiently evidenced by that precedent alone. Similarly, a recorded allegation is not automatically a judicial finding. These distinctions must be checked during annotation.

C. Evidence Selectors and Inference Rule

The production policy uses 190 token evidence windows with 40 token overlap and a maximum of 48 windows. If more candidates are available, semantic selection limits the set. Claims are capped at 140 tokens and NLI inputs at 384 tokens. All experimental variants use the same frozen NLI model and a threshold of 0.80. Model revisions are recorded in models/manifest.json. Table I describes the comparison.

TABLE I. EVIDENCE SELECTION POLICIES

Selector	Policy	Evidence allowance
Character prefix	Opening text in source order	1000 characters
Current windows	Production window selection	At most 48 windows
Budget prefix	Earliest candidate windows	380 tokens
Budget uniform	Endpoints then recursive midpoints	380 tokens
Budget semantic	Claim similarity order	380 tokens

For a claim c and selected windows W, let E(c) = max over w in W of p(entailment | w, c), and C(c) = max over w in W of p(contradiction | w, c). With threshold t = 0.80, the output is supported when E(c) >= t and C(c) < t; contradicted when C(c) >= t and E(c) < t; mixed when both exceed or equal t; and insufficient otherwise. These are model scores, not calibrated probabilities of legal truth.

For the budget variants, the total selected evidence allowance is B = 380 tokens. A final window may be shortened at a token boundary to fit the remaining allowance. Every real pilot example consumes exactly this budget under each of the three policies. Equal evidence tokens do not imply equal computation: pair counts, repeated claim tokens, padding, and semantic scoring overhead can differ. The character prefix and current window comparison intentionally retain unequal evidence exposure and is reported separately.

D. Datasets and Reproducibility

The synthetic set includes 54 constructed examples: six scenario families, three evidence positions, and three reference labels. Early, middle, and late placement stress tests whether selection omits decisive evidence. This is an engineered diagnostic set rather than a sample of authentic legal questions. Its labels are controlled by fixture labels, not legal expert annotations.

The real pilot includes six complete judgments from the project CSV and 30 constructed claims. Each judgment contributes two draft supported claims, two draft contradicted claims, and one claim concerning an unstated detail. Topics include property and adoption, tenancy, appellate tax powers, agricultural tax proceedings, a conviction appeal, and dividend taxation. Claims and labels were drafted after reading the source texts and frozen before model inference. The source text matches its CSV row apart from inherited leading and trailing whitespace trimming; passages were not inserted, moved, or rewritten. Court official originals were not independently authenticated.

The pilot is a convenience sample, not a held out benchmark. Draft labels are stored separately from null gold labels, preventing the unreviewed file from passing reviewed evaluation mode. Source snapshots, dataset hashes, per example predictions, and reference spans support auditability. Two independent reviewer sheets with neutral identifiers have been prepared, but review has not occurred. A final study must adjudicate disagreements and evaluate new cases because the present pilot outcomes have already been inspected.

E. Evaluation Measures

The primary real data measure is draft label agreement: the proportion of outputs matching the frozen draft label. Macro F1 averages the three reference class scores; mixed predictions count as errors. Reference span recovery is measured only for the 24 claims with annotated supporting or contradicting spans. Additional diagnostics include supported claim coverage, unsupported acceptance, mixed outputs, and latency. Recovering a reference span indicates exposure to the annotated text, not correct legal interpretation.

Paired differences use 2000 case group bootstrap resamples, keeping claims from the same judgment together. Synthetic resampling groups examples by scenario family. The reported intervals are exploratory and unadjusted for multiple comparisons. With only six groups, they do not resolve sampling or annotation uncertainty. Timing includes selection and NLI after warm up, excludes loading and input/output, and uses one observation per method and example. Execution used Windows with 16 logical processors and four PyTorch threads; hardware details are insufficient for a general performance claim.

IV. RESULTS AND DISCUSSION

A. Real Judgment Pilot

TABLE II. REAL PILOT RESULTS AGAINST AI ASSISTED DRAFT LABELS

Selector	Agreement	Macro F1	Span recovery	p50 seconds
Character prefix	12/30 40.0%	0.333	41.7%	0.128
Current windows	16/30 53.3%	0.449	100.0%	0.879
Budget prefix	10/30 33.3%	0.275	50.0%	0.131
Budget uniform	15/30 50.0%	0.421	58.3%	0.193
Budget semantic	15/30 50.0%	0.422	66.7%	0.411

Current windows improve draft agreement over the character prefix by 13.3 percentage points, with an exploratory 95% case bootstrap interval of [6.7, 20.0] points. However, they process an average of 2574.2 evidence tokens compared with 295.3 for the character prefix. This result cannot attribute the gain solely to window placement. At 380 tokens, semantic selection exceeds budget prefix agreement by 16.7 points, interval [10.0, 20.0], but ties uniform selection; the semantic minus uniform interval is [-10.0, 10.0]. The pilot provides no aggregate agreement advantage for semantic selection over a simpler uniform baseline.

The critical negative result is that none of the five methods returns supported for any of the 12 draft supported claims. Current windows recover all 24 annotated spans, yet produce contradiction for six draft supported claims, insufficient evidence for five, and mixed for one. Their agreement gain comes from recognizing 10 of 12 draft contradicted claims compared with six for the character prefix. Both methods match all six draft insufficient labels. Thus, improved coverage does not solve the verifier problem. Zero unsupported acceptance is uninformative as a safety success because supported acceptance is also zero.

B. Synthetic Position Stress Test

TABLE III. SYNTHETIC DIAGNOSTIC RESULTS ON 54 EXAMPLES

Selector	Correct	Accuracy	Macro F1
Character prefix	28/54	51.9%	0.504

Selector	Correct	Accuracy	Macro F1
Current windows	43/54	79.6%	0.788
Budget prefix	35/54	64.8%	0.646
Budget uniform	35/54	64.8%	0.637
Budget semantic	40/54	74.1%	0.736

The synthetic stress test exhibits the intended position effect. Character prefix accuracy is 100.0% for early evidence and 27.8% for both middle and late evidence. Current windows achieve 94.4%, 66.7%, and 77.8%, respectively. Windows therefore improve access to later evidence but are not uniformly better, even on these fixtures. Their overall difference from the character prefix is 27.8 points, interval [20.4, 37.0]. Under the 380 token constraint, semantic selection exceeds budget prefix and uniform selection by 9.3 points each, with intervals [0.0, 18.5] and [0.0, 20.4]. These engineered results motivate further tests; they do not establish real legal accuracy.

C. Interpretation and Threats to Validity

The evidence supports two bounded findings. First, opening only evidence selection can omit relevant later passages. Second, complete exposure to an annotated passage is insufficient for correct inference with the current verifier. The real data failure could reflect model domain mismatch, label ambiguity, attribution errors, or aggregation across conflicting statements. The present experiment does not isolate those causes. Future error analysis should distinguish allegations from findings, align entities and times, and test calibrated thresholds and alternative verifiers on development data before freezing a new evaluation.

The main limitations are small sample size, correlated claims, assistant authored draft labels, convenience case selection, and lack of official source authentication. Verification uses complete supplied judgments rather than retrieved evidence, and therefore omits retrieval failures. The production and prefix comparison changes both coverage and evidence volume. Latency is descriptive rather than a repeated hardware benchmark. The literature review is limited to the supplied 20 papers and cannot establish exhaustive novelty. These constraints prevent claims of legally validated accuracy, production readiness, or superiority over the reviewed complete systems.

The reproducibility package includes evaluate_verification.py, the annotation protocol, frozen pilot sources, reviewer forms, model revision records, and machine readable reports. The implementation checks recorded 36 passing tests and eight production parity checks. These checks establish software consistency within their tested scope; they do not validate draft legal labels. A conference submission should add independent adjudication, a larger new benchmark, and a specified venue template before making confirmatory claims.

V. CONCLUSION

The pilot shows that selecting more of a judgment can improve evidence coverage without producing reliable claim verification. Current windows raise agreement with draft labels from 40.0% to 53.3%, while semantic and uniform selection tie at 50.0% under the same evidence budget. However, every selector misses all draft supported claims. Failure is central to the finding: better access to text does not ensure that the model understands it. The study provides a repeatable way to examine that distinction. Independent legal review and a new held out benchmark are needed before drawing conclusions about legal accuracy.

REFERENCES

[1] Nicholas Pipitone, Ghita Houir Alami, “LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain,” arXiv preprint 2408.10343v1, 2024. https://arxiv.org/abs/2408.10343v1

[2] Minhu Park, Hongseok Oh, Eunkyung Choi, Wonseok Hwang, “LRAGE: Legal Retrieval Augmented Generation Evaluation Tool,” arXiv preprint 2504.01840v2, 2025. https://arxiv.org/abs/2504.01840v2

[3] Shubham Kumar Nigam et al., “AILQA: Evaluating AI-driven legal question answering systems for the indian legal system,” Artificial Intelligence and Law DOI: 10.1007/s10506-026-09537-2, 2026. https://doi.org/10.1007/s10506-026-09537-2

[4] Nirmalie Wiratunga et al., “CBR-RAG: Case-Based Reasoning for Retrieval Augmented Generation in LLMs for Legal Question Answering,” arXiv preprint 2404.04302v1, 2024. https://arxiv.org/abs/2404.04302v1

[5] Shatrunjay Kumar Singh, “Precedent-Aware Multi-Agent Retrieval-Augmented Generation in Case Law Analysis,” International Journal of Innovative Science and Research Technology 10(12), 2546-2550, 2025. https://doi.org/10.38124/ijisrt/25dec1556

[6] Zerui Chen et al., “LegalGraphRAG: Multi-Agent Graph Retrieval-Augmented Generation for Reliable Legal Reasoning,” ACL 2026 64th Annual Meeting, Volume 1: Long Papers 37455-37484, 2026. https://aclanthology.org/2026.acl-long.1738/

[7] Joy Bose, “Falkor-IRAC: Graph-Constrained Generation for Verified Legal Reasoning in Indian Judicial AI,” arXiv preprint 2605.14665v2, 2026. https://arxiv.org/abs/2605.14665v2

[8] Pardeep Kumar, Vijay Dhir, “Enhancing Legal Question Answering with Evidence-Grounded RAG, Multi-Model Evaluation, and Safe-Template Repair,” International Journal of Drug Delivery Technology (IJDDT) 16(59s), 1086-1093, 2026.

[9] Sayed Ayaan Ahmed Sha, Sangeetha Sivanesan, Anand Kumar Madasamy, Navya Binu, “Rhetorical-Role-Aware Retrieval-Augmented Generation for Legal Question Answering over Indian Supreme Court Judgments,” arXiv preprint 2608.06828v1, 2026. https://arxiv.org/abs/2608.06828v1

[10] Vishnuprabha V, Daleesha M Viswanathan, Rajesh R, Aneesh V. Pillai, “Augmented Question-guided Retrieval (AQgR) of Indian Case Law with LLM, RAG, and Structured Summaries,” arXiv preprint 2508.04710v1, 2025. https://arxiv.org/abs/2508.04710v1

[11] Rakshita Goel et al., “Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India,” arXiv preprint 2602.23371v1, 2025. https://arxiv.org/abs/2602.23371v1

[12] Galih Wasis Wicaksono et al., “Leveraging Retrieval-Augmented Generation to Support Legal Question Answering: Evidence from Indonesian Human Trafficking Court Decisions,” International Journal on Informatics Visualization 10(4), 1674-1683, 2026.

[13] Shubham Kumar Nigam et al., “NyayaRAG: Realistic Legal Judgment Prediction with RAG under the Indian Common Law System,” arXiv preprint 2508.00709v3, 2025. https://arxiv.org/abs/2508.00709v3

[14] Parjanya Aditya Shukla et al., “NyayaMind: A Framework for Transparent Legal Reasoning and Judgment Prediction in the Indian Legal System,” arXiv preprint 2604.09069v1, 2026. https://arxiv.org/abs/2604.09069v1

[15] Soham Sachin Shelar, Manisha Bharati, “LegalMind: A Multi-Agent Legal Reasoning Framework Leveraging Fine-Tuning of Large Language Models and Retrieval-Augmented Generation to Reduce Hallucinations,” International Journal of Innovative Science and Research Technology 11(5), 3431-3439, 2026. https://doi.org/10.38124/ijisrt/26may1714

[16] Mustipalli Naveen, Komera Ashok, Madana Akash Reddy, Madisetty Gowrinadh, “Multi-Agent Retrieval Augmented Generation System for Legal Applications: A Neuro-Symbolic Approach,” International Journal for Research in Applied Science & Engineering Technology 14(IV) DOI: 10.22214/ijraset.2026.78630, 2026. https://doi.org/10.22214/ijraset.2026.78630

[17] Deepanshu et al., “NyayaAI: An AI-Powered Legal Assistant Using Multi-Agent Architecture and Retrieval-Augmented Generation,” arXiv preprint 2605.10155v1, 2026. https://arxiv.org/abs/2605.10155v1

[18] Parag Patil et al., “Design and Implementation of Legal Gennie: A Retrieval-Augmented Multilingual Legal Advisory System,” International Journal of Scientific Research in Computer Science, Engineering and Information Technology 12(2), 271-277, 2026.

[19] Karthik Surya N A, Vedabhishekh T, Vidhya Muthulakshimi, Purushothaman G, “Towards Intelligent Legal Information Retrieval a Transformer Based Framework,” International Journal of Scientific Research in Engineering and Management 9(5) DOI: 10.55041/IJSREM48260, 2025. https://doi.org/10.55041/IJSREM48260

[20] Manjunath S et al., “Vakil - A Virtual Assistant for Knowledge in Indian Law,” International Journal of Scientific Research in Engineering and Management 9(12) DOI: 10.55041/IJSREM54798, 2025. https://doi.org/10.55041/IJSREM54798